

AvatarFit

Formal User Test + SUS — Results

HCI Evaluation

1. Overview

Three participants took part in the formal usability test of AvatarFit. Sessions were conducted in-person with a single facilitator. Each session lasted approximately 20 minutes and followed a structured protocol covering four tasks: character selection, intentional bad form (squat), corrected form (squat), and lunge (bad then corrected). At the end of each session, participants completed the System Usability Scale (SUS) questionnaire. This evaluation took place after a design iteration cycle in which three issues identified during the Week 2 Think-Aloud pilot were resolved: a female avatar was added, a side-orientation cue was introduced for squat and lunge, and voice feedback was expanded to include full verbal corrections.

Session summary:

Participant	Initials	SUS Score	Category
Participant 1	A.Z.U	82.5 / 100	Excellent
Participant 2	S.G.C	75.0 / 100	Good
Participant 3	A.S.S	82.5 / 100	Excellent

2. Task Observations

Observations recorded by the facilitator during each structured task.

Task 1 — Character Selection and Exercise Start

Observation	A.Z.U	S.G.C	A.S.S
First impression / comment	Liked the interface	No use problem noted	No use problem noted
Finds character selection	Immediately	Immediately	Immediately
Finds exercise pill	Immediately	Immediately	Immediately

Task 2 — Intentional Bad Form (Squat)

Observation	A.Z.U	S.G.C	A.S.S
Time to first feedback	2-3 seconds	2-3 seconds	2-3 seconds
Comment / reaction	Surprised the app caught wrong movements so easily	No comment; no use problem	Found the recommendation useful

Task 3 — Corrected Form (Squat)

Observation	A.Z.U	S.G.C	A.S.S
Comment / reaction	Understood the correction without any problem	At first did not follow instructions correctly; after a few tries skeleton turned green	Movement corrected at first try

Task 4 — Switch to Lunge (Bad then Corrected)

Observation	A.Z.U	S.G.C	A.S.S
Overall outcome	Exercise completed successfully	Exercise completed successfully	Exercise completed successfully

3. Final Comments

A.Z.U	<i>"I found the app very original and interesting; I think that the spoken feedback was very useful."</i>
S.G.C	<i>"I usually don't do a lot of exercise, but I think that the app was really useful and I will definitely use it at home."</i>
A.S.S	<i>"I like it a lot. I think the avatar was very cute, and in general the app was well-organised and well-executed."</i>

4. SUS Results

Each participant completed the 10-item System Usability Scale after the test session. Odd-numbered items are scored as (response - 1); even-numbered items as (5 - response). The sum of all 10 adjusted scores is multiplied by 2.5 to give a final score out of 100.

#	Statement	P1 raw	P1 score	P2 raw	P2 score	P3 raw	P3 score
1	I think that I would like to use this app frequently.	4	3	3	2	5	4
2	I found the app unnecessarily complex.	2	3	2	3	2	3
3	I thought the app was easy to use.	4	3	4	3	4	3
4	I think that I would need the support of a technical person to use this app.	2	3	2	3	2	3
5	I found the various functions in this app were well integrated.	5	4	4	3	4	3
6	I thought there was too much inconsistency in this app.	2	3	1	4	2	3
7	I would imagine that most people would learn to use this app very quickly.	4	3	3	2	4	3
8	I found the app very cumbersome to use.	1	4	2	3	2	3
9	I felt very confident using the app.	4	3	4	3	5	4
10	I needed to learn a lot of things before I could get going with this app.	1	4	1	4	1	4
Sum of adjusted scores		—	33	—	30	—	33
SUS Score (sum × 2.5)		82.5	Excellent	75.0	Good	82.5	Excellent

Mean SUS Score: **80.0 / 100** — *Good / Excellent*

< 51 Poor	51–67 OK	68–80 Good	81–100 Excellent	Our result: 80 Good / Excellent
			◀ <i>our result</i>	

5. Key Findings

OK All tasks completed successfully by all participants Every participant completed all four tasks (character selection, bad squat, corrected squat, lunge) without requiring facilitator intervention. Task completion rate: 100%.

OK Bad form detection is fast and noticeable All three participants received visual and audio feedback within 2-3 seconds of performing incorrect form. One participant spontaneously commented on how quickly the app caught mistakes.

OK Corrective feedback is now clear and actionable Two out of three participants corrected their squat form on the first attempt, and the third succeeded after a few tries. The voice feedback improvements made after the Think-Aloud appear to have resolved the main issue identified in that test.

OK SUS score of 80/100 — Good / Excellent The mean SUS score places AvatarFit at the boundary between 'Good' and 'Excellent' usability. All three individual scores (82.5, 75, 82.5) exceed the 68-point threshold for above-average usability.

(!) Minor learning curve on corrected squat remains One participant (S.G.C) needed several attempts to achieve the correct squat posture. This suggests the corrective feedback, while improved, may still benefit from additional specificity.

OK High self-reported likelihood of home use All three participants expressed in their final comments that they would use the app independently at home, validating the core HCI problem statement from the first survey.

6. Conclusions

The formal user test confirms that the design iterations implemented after the Think-Aloud evaluation were effective. All three participants completed every task successfully, bad form was detected within 2-3 seconds across all sessions, and the expanded voice feedback was explicitly praised by one participant. The mean SUS score of 80/100 places AvatarFit above the threshold for good usability and close to the excellent range.

The one remaining minor issue — a slight learning curve on corrected squat form for one participant — does not represent a critical usability failure but suggests that further refinement of corrective feedback specificity could push the SUS score into the excellent band in future iterations.

Most significantly, all three participants stated that they would use AvatarFit at home for independent workouts, directly validating the original HCI design goal: providing accessible, real-time exercise form correction for users who work out without a personal trainer.